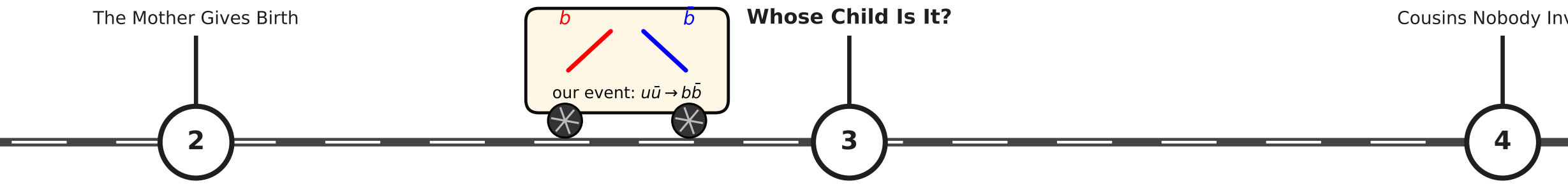
Stop 3: Whose Child Is It? Matching and merging

This stop is for events with

a second parent for the extra parton

- $b\bar{b} + 1$, 2-parton subprocesses: MLM, CKKW-L merging
- a next-to-leading-order matrix element: MC@NLO, POWHEG matching
- both at once: FxFx



Chapter 3 is a stop on the road: the sign says who it is for.

Stop 3: Whose Child Is It? Matching and merging

Our event

- one leading-order $2 \to 2$ process: $u\bar{u} \to b\bar{b}$
- no $u\bar{u} \to b\bar{b}g$, no $ug \to b\bar{b}u$ in the sample
- every extra parton has one parent, the shower
- nothing to match, nothing to merge

NOT FOR US

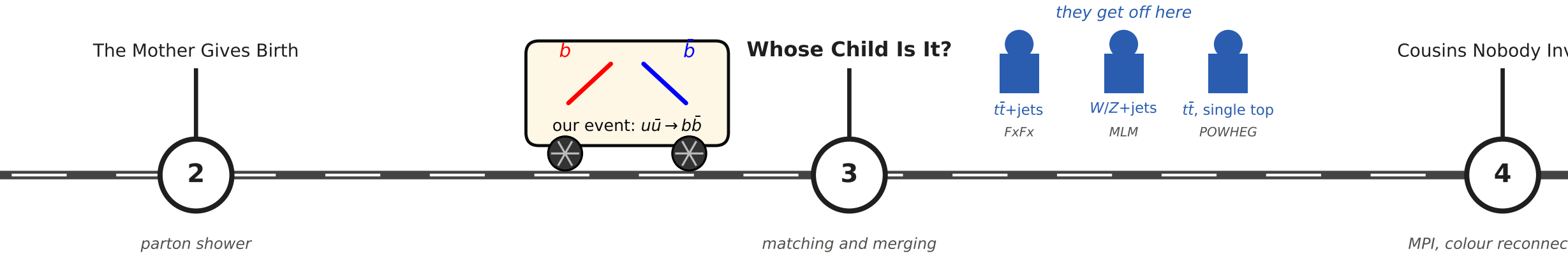
ledger

b family after the shower	100.45 GeV, $m = 13.52$ GeV, 3 partons
after matching and merging (none applied)	**100.45 GeV,** $\Delta p_T = 0$

This stop is for events with

a second parent for the extra parton

- $b\bar{b} + 1$, 2-parton subprocesses: MLM, CKKW-L merging
- a next-to-leading-order matrix element: MC@NLO, POWHEG matching
- both at once: FxFx



Our event has one parent for every parton. There is nothing to match here, and the ledger row repeats.

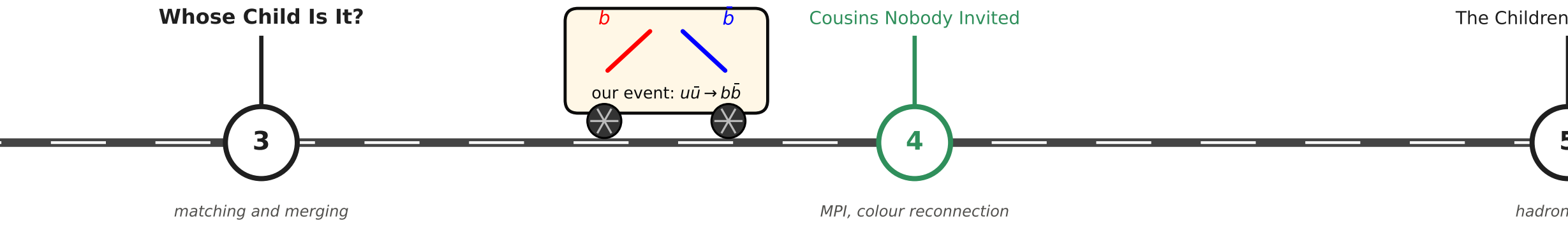
Stop 3: Whose Child Is It? Matching and merging

no stop here; on to the next one

This stop is for events with

a second parent for the extra parton

- $b\bar{b} + 1$, 2-parton subprocesses: MLM, CKKW-L merging
- a next-to-leading-order matrix element: MC@NLO, POWHEG matching
- both at once: FxFx



So we roll on: chapter 4, the rest of the two protons.